

The Regime-Consistent Measurement Principle (RCMP): A Methodological Framework for Multi-Scale Physics

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Abstract

We introduce the Regime-Consistent Measurement Principle (RCMP), a methodological framework designed to prevent interpretive errors when measurements span multiple physical regimes. The principle states that an observable must be interpreted through the variable most directly coupled to the physical driver in the phenomenon’s operative regime, with cross-regime mappings requiring explicit transformation and uncertainty propagation. We formalize five core principles: driver proximity, regime tagging, proxy accounting, coordinate humility, and cross-validation. The framework is demonstrated through application to galaxy rotation curve analysis, where it successfully distinguishes between per-point scatter (intra-galaxy variability) and per-galaxy median response (inter-galaxy physics), resolving apparent discrepancies in the low-acceleration regime. RCMP provides a general methodology applicable across astrophysics, cosmology, and any domain where measurements must be interpreted across different physical scales or regimes.

Keywords: measurement methodology, regime validity, proxy variables, galactic dynamics, radial acceleration relation, epistemology of physics

1 Introduction

Modern physics increasingly confronts phenomena that span multiple physical regimes—from quantum to classical, from laboratory to cosmological scales, from high to low acceleration environments. A persistent challenge in such multi-regime contexts is the interpretation of measurements: which variables should be considered primary, and how should transformations between different descriptive frameworks be handled?

Consider galaxy rotation curves. The same kinematic data (observed velocities) can be interpreted through multiple theoretical lenses: mass distributions in Λ CDM, acceleration scales in MOND/RAR, or energy flow in alternative frameworks. Each interpretation

implicitly assumes certain variables are “primary” and others are “derived.” When these assumptions are not made explicit, apparent disagreements between frameworks may reflect methodological artifacts rather than physical differences.

We propose the **Regime-Consistent Measurement Principle (RCMP)** as a methodological guard against such regime-mixing errors. The principle does not adjudicate between physical theories but provides a framework for ensuring that measurements are interpreted consistently within their operative regime.

2 Statement of the Principle

2.1 Formal Definition

RCMP (English): An observable must be interpreted through the variable most directly coupled to the physical driver in the phenomenon’s operative regime. Cross-regime mappings require explicit transformation and uncertainty propagation.

RCMP (Norwegian): En observabel skal tolkes gjennom den variabelen som er mest direkte koblet til den fysiske driveren i fenomenets operative regime. Kryss-regime-mapping krever eksplisitt transformasjon og usikkerhetsføring.

2.2 Core Principles

The RCMP operationalizes through five core principles:

1. **Driver Proximity:** Choose the most directly coupled observable within the operative regime. In galaxy dynamics, if gravity is the driver, use acceleration-based variables rather than mass-based proxies that require additional modeling assumptions.
2. **Regime Tagging:** Each measurement must be labeled with its dominant regime(s). We adopt a four-level epistemic hierarchy (L0–L3) ranging from directly measured quantities to highly derived theoretical constructs.
3. **Proxy Accounting:** Each transformation from measurement to interpretation carries an explicit uncertainty budget entry. The cumulative proxy chain—from raw data through calibrations, model assumptions, and theoretical mappings—must be documented.
4. **Coordinate Humility:** Global coordinates may be background (fixed reference) in some regimes but dynamic (physically meaningful) in others. The choice of coordinate system is not epistemically neutral.
5. **Cross-Validation:** Consistency must be tested across different proxy chains. If two independent measurement pathways yield the same physical conclusion, confidence increases; if they diverge, the divergence itself is informative.

3 Theoretical Foundation

3.1 Epistemic Layer Structure (L0–L3)

RCMP extends naturally from hierarchical epistemic frameworks. We adopt a four-layer structure:

Layer	Description	Example (Galaxy Dynamics)
L0	Direct measurement	Spectral line velocity
L1	Calibrated observable	Rotation velocity $V(R)$
L2	Derived quantity	Centripetal acceleration g_{obs}
L3	Theoretical construct	Dark matter density profile

Table 1: Epistemic layer structure with galaxy dynamics examples.

The principle requires that interpretive claims at layer L_n be traceable to measurements at L0, with explicit transformation rules and uncertainty propagation at each step.

3.2 Regime-Dependent Validity

Physical theories and models have regimes of validity—domains where their assumptions hold and their predictions are reliable. RCMP formalizes the requirement that measurements be interpreted within the appropriate regime:

- A measurement taken in regime A should not be directly compared to predictions calibrated in regime B without explicit transformation.
- When a phenomenon spans multiple regimes (e.g., galaxy rotation curves spanning high and low acceleration), the analysis must respect regime boundaries.

3.3 Relationship to Existing Frameworks

RCMP is consistent with, but distinct from:

- **Uncertainty quantification:** RCMP goes beyond error bars to require explicit proxy chain documentation. It complements uncertainty quantification by structuring where uncertainties originate along the proxy chain.
- **Model selection criteria:** RCMP is pre-theoretical; it applies before model comparison.
- **Dimensional analysis:** RCMP addresses interpretive choices, not just unit consistency.

4 Application: Galaxy Rotation Curves

4.1 The Problem

Galaxy rotation curves exhibit a tight relationship between observed centripetal acceleration (g_{obs}) and that predicted from baryonic matter (g_{bar})—the Radial Acceleration Relation (RAR) (McGaugh et al., 2016). However, scatter in this relation varies across the acceleration range, with notably higher scatter at low accelerations ($g_{\text{bar}} < 10^{-11} \text{ m/s}^2$), particularly in dwarf galaxies.

Interpretations of this scatter include:

- Measurement systematics (distance errors, inclination uncertainties)
- Kinematic effects (non-circular motions, pressure support)
- Intrinsic physics (additional degrees of freedom in the response law)

Without RCMP, these interpretations are often conflated.

4.2 RCMP-Guided Analysis

Applying RCMP to this problem requires:

1. **Driver identification:** In the low-acceleration regime, the driver is the gravitational response to baryonic mass.
2. **Regime tagging:** Data points are tagged by acceleration regime (high- g vs. low- g) and data quality (Q-flag).
3. **Proxy accounting:** The chain from observed velocity to inferred acceleration includes:
 - Distance (affects R and mass calibration)
 - Inclination (affects velocity projection)
 - Mass-to-light ratios (affects g_{bar})
4. **Analysis structure:** Separate per-point analysis (all data points) from per-galaxy analysis (median residuals per galaxy).

4.3 Results Summary

RCMP-guided analysis of the SPARC dataset (Lelli et al., 2017) reveals:

Finding	Per-Point	Per-Galaxy
Dwarf vs spiral scatter	1.27 $\times$ higher ($p = 0.02$)	Identical ($p = 0.47$)
Interpretation	Intra-galaxy variability	No class-level offset

Table 2: Per-point vs per-galaxy analysis distinguishes intra-galaxy variability from inter-galaxy physics.

This distinction resolves the apparent paradox: dwarf galaxies show more internal variability (consistent with kinematic or observational factors), but their median response follows the same law as spirals (consistent with universal physics).

Further tests of specific systematics hypotheses:

Hypothesis	Proxy Tested	Result
High-frequency kinematic noise	Roughness ratio	Not supported
Outer-disk systematic	Radius trend	Not supported
Pressure support	V_{flat} correlation	Wrong sign
Distance uncertainty	e_D correlation	Supported (Holm $p = 0.035$)

Table 3: Systematic hypothesis tests with multiplicity correction.

4.4 Epistemological Status

After RCMP-guided analysis, we can state:

“The low-acceleration manifold is real at the galaxy level. Excess scatter in dwarfs is internal and low-frequency. Distance uncertainty explains some amplitude variation. High-frequency kinematic and simple pressure-support explanations are disfavored. A weak V_{flat} anomaly remains unexplained.”

This is a **problem definition**, not a conclusion. It maps the boundary between what is explained and what remains open—exactly what RCMP is designed to achieve.

5 Generalization

5.1 Applicability Beyond Galaxy Dynamics

RCMP applies wherever measurements must be interpreted across regimes:

- **Cosmology:** CMB anisotropies interpreted through different epochs
- **High-energy physics:** Particle properties measured at different energy scales
- **Climate science:** Local measurements interpreted through global models
- **Neuroscience:** Single-neuron recordings interpreted through network models

5.2 Relationship to Theory Choice

RCMP is explicitly **pre-theoretical**. It does not favor Λ CDM over MOND, or any alternative framework over standard cosmology. Instead, it ensures that whatever theoretical framework is applied, the measurement interpretation is self-consistent.

Different theories may disagree about which variable is “most directly coupled to the driver”—and this disagreement is itself informative. RCMP makes such disagreements explicit rather than hidden. Importantly, RCMP may lead different theories to prioritize different primary variables; such divergence is itself a testable difference.

5.3 Connection to Multi-Model Philosophy

The principle supports the view that different models may be valid descriptions in different regimes:

“All models are ways of seeing the same system—they describe different coordinates on the same phenomenon, not different universes.”

RCMP operationalizes this philosophy by requiring explicit regime boundaries and transformation rules.

6 Formal Specification

6.1 Notation

Let:

- O = raw observable
- V_i = candidate interpretive variable ($i = 1, \dots, n$)
- D = physical driver in regime R
- $d(V_i, D | R)$ = epistemic distance from V_i to D in regime R
- T_{ij} = transformation from V_i to V_j
- $\sigma_{T_{ij}}$ = uncertainty introduced by transformation

6.2 RCMP Formal Statement

For observable O in regime R with driver D :

1. **Primary variable selection:**

$$V^* = \arg \min_i d(V_i, D | R) \quad (1)$$

2. **Transformation requirement:** For any $V_j \neq V^*$, interpretation requires:

- Explicit $T_{*j} : V^* \rightarrow V_j$
- Propagated uncertainty $\sigma_{\text{total}} = \sqrt{\sigma_O^2 + \sigma_{T_{*j}}^2}$

3. **Cross-regime mapping:** For regime $R' \neq R$:

- Identify D' and V'^* in R'
- Document regime boundary conditions
- Apply inter-regime transformation with explicit assumptions

Item	Requirement	Status
Driver identified	Physical driver explicit for each regime	☐
Regime tagged	Each data point labeled with dominant regime	☐
Proxy chain documented	Full transformation sequence recorded	☐
Uncertainties propagated	Each transformation adds to error budget	☐
Cross-validation performed	Multiple proxy chains compared	☐

Table 4: RCMP validation checklist.

6.3 Validation Checklist

7 Discussion

7.1 Limitations

RCMP requires judgment about:

- What constitutes a “physical driver” (theory-dependent)
- How to measure “epistemic distance” (operationalization needed)
- Where regime boundaries lie (may be gradual rather than sharp)

These are features, not bugs: RCMP makes these judgments explicit rather than implicit.

7.2 Relationship to Falsificationism

RCMP increases falsifiability by:

- Making measurement-to-driver coupling explicit
- Requiring predictions to specify their operative regime
- Enabling targeted tests of specific proxy chain steps

A theory that fails RCMP validation (e.g., makes claims outside its proxy chain’s validity) is more clearly falsified than one where the failure might be attributed to unexamined assumptions.

7.3 Future Development

Extensions in development include:

- Hierarchical Bayesian formalization of proxy accounting
- Automated regime classification from data characteristics
- Cross-domain validation (testing RCMP in quantum vs. classical boundaries)

8 Conclusion

The Regime-Consistent Measurement Principle provides a methodological framework for interpreting measurements that span multiple physical regimes. By requiring explicit identification of physical drivers, systematic regime tagging, documented proxy chains, coordinate humility, and cross-validation, RCMP prevents interpretive errors that arise from implicit regime-mixing.

Application to galaxy rotation curves demonstrates the principle’s utility: apparent discrepancies between galaxy classes resolve into distinct components (intra-galaxy variability vs. inter-galaxy physics) that require different explanations. This is not a physical conclusion but an epistemic clarification—exactly what methodology should provide.

RCMP does not replace physical theories; it clarifies how they are to be tested.

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Data Availability

Analysis scripts and results are available at: [Repository URL]

The SPARC dataset is publicly available at: <http://astroweb.cwru.edu/SPARC/>

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A Worked Example — Low- g Scatter Analysis

A.1 Initial Observation

Scatter in RAR residuals increases at low accelerations. Dwarfs ($T \geq 8$) show 27% higher scatter than spirals in $Q=1$ data.

A.2 RCMP-Guided Decomposition

Step 1: Regime identification

- Low- g regime: $g_{\text{bar}} < 10^{-11} \text{ m/s}^2$
- Quality filter: $Q = 1$ (highest quality)

Step 2: Level separation

- Per-point: tests for internal structure
- Per-galaxy: tests for class-level physics

Step 3: Proxy chain analysis

- Tested: Inc, e_{Inc} , Distance, e_D , V_{flat}
- Each correlation computed with multiplicity correction

A.3 Results

Test	Holm-adjusted p	Interpretation
$e_D \rightarrow$ amplitude	0.035	Distance uncertainty contributes
$V_{\text{flat}} \rightarrow$ median	0.053 (FDR)	Unexplained anomaly
Roughness ratio	> 0.15	No high-frequency excess

Table 5: Summary of systematic hypothesis tests.

A.4 Conclusion

Per RCMP, we can state: “Distance uncertainty robustly predicts amplitude scatter; kinematic proxies do not support high-frequency systematics; one anomaly ($V_{\text{flat}}-\mu$ correlation) remains unexplained.”

This is a well-characterized problem definition, not a resolved question.